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Cultural resilience of heritage sites: Dimension exploration and scale development

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ABSTRACT

Heritage sites are in a state of instability, facing various disturbances or environmental changes, which may lead to a crisis in cultural protection and inheritance. Importantly, cultural resilience is the key to dissipating these disturbances or even transforming them into sustainable development opportunities, but few studies have conducted theoretical exploration and evaluation on this topic. Therefore, this study aims to conceptualize and develop scales for cultural resilience of heritage sites (CRHS) using online and offline investigations (in-depth interviews, big data mining, and questionnaires). The results from grounded theory procedures reveal that cultural resilience consists of 5 dimensions: cultural resistance, restoration, learning ability, adaptability, and growth. Moreover, a cultural resilience scale with 17 items was developed and proved reliable and valid. This study lays an empirical foundation for the theoretical system development of CRHS and provides resilience ideas and strategies for their cultural protection, inheritance, and sustainable development.

1. Introduction

Heritage sites utilize cultural heritage resources to attract tourists, and there are also high requirements for their cultural protection and inheritance (Bui et al., 2020; Seyfi et al., 2019). Tourism development can empower cultural inheritance through cultural dissemination and economic value creation, but it also brings harmful interference, such as cultural conflict and overcommercialization (Ghermandi et al., 2020; Hu et al., 2021). Therefore, the traditional culture of heritage sites is more likely to encounter various internal and external impacts during their development, resulting in negative issues, i.e., cultural distortion and cultural transplantation (Zhang et al., 2023). These issues will damage cultural authenticity and integrity and trigger a cultural inheritance crisis (Li et al., 2022; Pesce et al., 2019). As a critical category for cultural adaptation to environmental changes and crisis impacts in heritage sites, cultural resilience can effectively reduce risk infringement and even turn a "crisis" into an "opportunity". It contributes to the sustainable development of heritage sites (Clarke & Mayer, 2017; Holtorf, 2018; Hu et al., 2021). However, most heritage sites regard cultural resilience only as a slogan to cater to cultural sustainability and still lack a clear understanding of "*what it is*". Therefore, it is urgent to clarify the connotation and structure of cultural resilience of heritage sites (CRHS) and to monitor the state of cultural heritage in response to disturbances or changes by developing a scale.

The importance of cultural resilience for cultural protection and the sustainable development of heritage sites has been noted (Hu et al., 2021). However, previous studies have primarily focused on disturbances and optimization strategies in the cultural protection and inheritance of heritage sites (Park et al., 2019; Szromek et al., 2021; Tolkach & Pratt, 2021). The profound resilience logic behind the ability to achieve cultural protection and inheritance in response to disturbances or changes has been less explored by scholars. Although Bui et al. (2020) attempted to analyze the resilience of cultural heritage in a tourism destination, their study focused on community resilience amid spatial and cultural changes induced by mass tourism. Targeted theoretical exploration of CRHS is still vacant, making it challenging to provide resilience ideas and theoretical guidance for cultural protection and inheritance, and sustainable development of heritage sites. Importantly, Holtorf (2018) emphasized that cultural resilience is crucial for heritage sites to accept uncertainty, absorb adversity, cope with change, and continue to make significant progress. While some studies have introduced the concept of cultural resilience and delved into its

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1447-6770/© 2024 The Authors. Published by Elsevier Ltd. on behalf of CAUTHE - COUNCIL FOR AUSTRALASIAN TOURISM AND HOSPITALITY EDUCATION. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

exploration, they primarily highlight the role of culture in community resilience, and focus more on aspects like customs and beliefs (Clarke & Mayer, 2017; Sakakibara, 2017). These studies concentrate on the utility of cultural resources after communities experience disturbances, rather than the cultural system as the disturbed subject (Beel et al., 2017; Wu & Yuan, 2023). Consequently, it is difficult to provide theoretical guidance for the successful response of cultural systems of heritage sites to shocks.

To date, preliminary progress has been made in measuring cultural resilience. However, current research has developed an indicator system in which it is only regarded as a subdimension or subsystem of community resilience or system resilience in tourism destinations (Bui et al., 2020; Hu et al., 2021; Zhang et al., 2022; Zhu et al., 2022). However, these assessment approaches have significant limitations in measuring CRHS. On the one hand, heritage site culture is an independent complex concept involving multiple elements (Wang et al., 2023). Its resilience system also includes a multidimensional structure, which differs from tourism destinations' other forms of resilience (Clarke & Mayer, 2017). Nevertheless, the above measurements only regard cultural resilience as a secondary system and directly assess it using statistical indicators. These measurements lack a targeted theoretical deconstruction of CRHS, hindering subsequent quantitative research and practical dilemma resolution. For example, Hu et al. (2021) developed an evaluation system for the resilience of destinations like cultural heritage sites, using seven indicators to measure the secondary subsystem of socio-cultural resilience, i.e., population density, tourist density, and grade of intangible culture heritage inheritors. Notably, the indicators cover a wide range of content, and only a few are directly related to cultural heritage elements, which may not effectively measure CRHS.

On the other hand, the measurement indicators of cultural resilience in different resilience assessment frameworks of previous tourism destinations vary significantly, resulting in their weak universality (Hu et al., 2021; Zhu et al., 2022). There are differences in the cultural elements covered by the cultural systems of different heritage sites (Yu & Xu, 2019), and cultural heritage evolves and develops over time (Holtorf, 2018). Therefore, these quantitative statistical indicators designed for different types of destinations (e.g., the number of cultural commercial streets and A-level scenic spots) have limited reference value. Haque and Hostetler (2021) also questioned the resilience measurement based on numerical indicator data. Although this method is relatively objective compared to survey questionnaires, given the complexity and dynamics of resilience (Prayag, 2023; Weis et al., 2021), these limited objective indicators can hardly encompass the full scope of actual resilience, highlighting their applicability limitations in different research contexts. In summary, current achievements have struggled to provide a valid assessment tool for the cultural resilience of heritage sites, which requires urgent discussion of CRHS dimensional exploration and scale development.

To address these limitations, this study aims to: (1) consider the cultural resilience of heritage sites as an independent system and identify its essential connotation and dimensional structure; (2) develop a targeted and scientific CRHS scale to capture the cultural resilience state based on the perceptions of multi-stakeholders; (3) rigorously test the reliability and validity of the CRHS scale. Theoretically, this study clarifies the structural dimensions and measurement tool for the cultural resilience of heritage sites. It advances abstract and complex research on the cultural resilience of tourism destinations to a decomposable and measurable stage, laying the foundation for subsequent empirical research on CRHS. Practically, the cultural resilience scale contributes to managers of heritage sites in proposing resilience programs and strategies for cultural inheritance and sustainable development.

2. Literature review and concept definition

2.1. Cultural resilience

Early resilience emerged and was described as the ability of a material to recover its original shape after deformation under external forces, while medium-term ecosystem resilience was defined as the ability of a system to return to its original stable state after a disturbance (Gunderson & Holling, 2002; Weis et al., 2021). Since the 1990s, resilience research expanded to the field of social-ecological systems, giving rise to evolutionary resilience perspectives. Resilience is considered dynamic, multidimensional, and multiscale (Gunderson & Holling, 2002; Prayag, 2023). It emphasizes the process of "bouncing forward," including absorption, learning, adaptation, and transformation (McAreavey, 2022). With the in-depth exploration of socio-ecological or community resilience, the importance of cultural resources in resilience building has become increasingly prominent, attracting greater attention to the concept of cultural resilience (Beel et al., 2017; Crane, 2010). On the one hand, cultural resilience is incorporated into the resilience system of rural communities, working together with other types of resilience (e.g., ecological and economic resilience). It can motivate rural areas to respond to crises or emergencies through resistance, adaptability, and transformation (Davis et al., 2021; Wu & Yuan, 2023). On the other hand, some studies analyze resilience from a cultural heritage perspective, emphasizing that building such resilience contributes to the stable development of cultural heritage communities (Beel et al., 2017; Choi et al., 2021; Khan et al., 2021).

Culture is primarily viewed as a powerful resource available to communities and individuals to cope with crises or challenges, representing a collection of customs, beliefs, and values (Clarke & Mayer, 2017; Sakakibara, 2017). Cultural resilience is also defined as the ability of communities to mobilize cultural resources to address external crises and threats (Clarke & Mayer, 2017). The lifestyles of community residents reflect the cultural resilience, demonstrating their positive adaptation process to cultural erosion caused by disturbances such as environmental factors (Sakakibara, 2017). Notably, Holtorf (2018) proposed that cultural resilience refers to the ability of cultural systems (composed of relevant cultural elements of local communities) to absorb adversity, cope with change, and continue to thrive. Some scholars have also introduced the concept of socio-cultural resilience, understanding it as how to readjust social and cultural structures after disturbances, maintain community stability, and devise new structures and norms that adapt to changes (Wu & Yuan, 2023).

Overall, these studies emphasize that cultural resilience is a crucial asset for communities or socioecological systems to deal with shocks or crises (Haque & Hostetler, 2021; Weis et al., 2021). In this context, their definitions of cultural resilience are not fully applicable to the cultural resilience of heritage sites, as they differ in terms of disturbed subject, specific composition of culture, role of culture, triggering situations, and manifestations (Fig. 1). Furthermore, there is insufficient research on the composition and measurement of cultural resilience.

In addition to being a resource, destination cultural systems are susceptible to various disturbances (Ghermandi et al., 2020; Hu et al., 2021). This issue is especially prominent for cultural heritage sites due to their high dependence on cultural resources for socio-economic development, local cultural protection, and inheritance, often subject to long-term disturbances or shocks (Hu et al., 2021; Yu & Xu, 2019). Therefore, it is urgent to seek targeted resilience solutions for cultural systems rather than merely focusing on the role of culture in community or socio-ecological system resilience. Notably, Prayag (2023) also suggested that any resilience assessment and construction should be grounded in the corresponding adversities faced by the system. Consequently, this study moves beyond previous studies that emphasize viewing culture as an essential resource for resilience building. It directly addresses the disturbances faced by heritage sites' cultural systems and explores the resilience logic behind their successful responses.

2.2. Definition of cultural resilience of heritage sites

Resilience is closely related to change and sustainable development.

Concept	Disturbed subject	Specific composition of culture	Role of culture	Triggering situations	Manifestations
Cultural resilience	Community or socioecologic-al systems, co-existing with culture	A collection of practices, beliefs, values, etc.	Resources for addressing disturbance: communities can use cultural resources to address shocks and adapt to changes	After the community or social-ecologic-al system is disturbed	Emphasizing coping with disturbances, partly involving absorbing advers-ity and continuing development
Cultural resilience of heritage sites	Cultural systems of heritage sites	Various tangible (e.g. ancient buildings) and intangible (e.g. folklore or traditional theater) heritage elements	Disturbed subject: the cultural system of heritage sites needs to utilize internal and external resources to address disturbances	After the cultural system of heritage sites undergoes shocks or changes during the tourism development	Including the process for coping with disturbances such as survival, adaptation, recovery and continued renewal and development

Fig. 1. The difference between the concept connotation of cultural resilience and CRHS.

As heritage sites undergo transformation and development amidst changes or disturbances, their cultural resilience has become increasingly prominent (Hu et al., 2021). Previously, cultural resilience has often been explored within community or socioecological system resilience frameworks. It is inclined to dynamic resilience (or evolutionary resilience) and is rooted in the complexity, uncertainty, shock, nonlinearity, and evolution of system development (Folke, 2006; Prayag, 2023). Resilience in this context also highlights the ability of systems to 'thrive' while facing challenges and extreme events (Magis, 2010; O'Brien, 2012). Scholars also support the notion that cultural resilience implies adaptation and transformation of cultural systems rather than necessarily returning to a previous undisturbed state (Bee et al., 2017; Bui et al., 2020).

In the context of tourism development, cultural heritage evolution is characterized by both dynamism and continuity. The cultural significance of heritage is not eternal but depends on the role it plays in society and needs to adapt to changing circumstances over time (Bui et al., 2020; Hu et al., 2021). Additionally, some aspects of cultural heritage are deeply rooted and continue throughout the development of tourism destinations, connecting the past, future, and present (Yu & Xu, 2019). Based on these characteristics, cultural resilience of heritage sites also implies continuity and change. It helps provide opportunities for cultural transformation or reorganization, contributing to cultural sustainability in a long-term embrace of uncertainty and constant renewal (Holtorf, 2018). CRHS encompasses the processes by which a system responds to change, whether facing shock, surprise, or more gradual change (McAreavey, 2022). In addition to resisting disruption, it includes coping pathways of adaptation, learning, self-organization, and continued development (Li et al., 2021). In summary, with a primary reference to Holtorf (2018), this study proposes that CRHS emphasizes its dynamic development ability. It refers to the ability of a destination's cultural system to survive, adapt, recover, and even continue to renew and develop while coping with disturbances or changes during heritage tourism development.

2.3. Resilience evaluation of tourism destinations

Tourism destinations are composed of multiple systems of culture, economy, society, technology, and physical environment (Pyke et al., 2021), and scholars have evaluated tourism destination resilience from these different aspects. Destination resilience and community resilience have received much attention in tourism research, and current references have regarded tourism destinations as complex systems, constructed a tourism destination resilience model, and have constantly improved and adjusted it (Dai et al., 2019; Guo et al., 2018; Pyke et al., 2021). In addition, community resilience and social-ecological system resilience include elements of economic resilience, ecological resilience, and cultural resilience. Some studies have built measurement index systems or evaluation items based on these elements (Holladay & Powell, 2013; Hu et al., 2021; Jamaliah & Powell, 2018). Moreover, different standard layers were also developed to evaluate these resilience, such as attraction, adaptability, and transformation (Jia et al., 2022; Yang et al., 2021). With the deepening of research, scholars have also separated tourism destinations' economic and social resilience and explored a more detailed framework of resilience evaluation index systems (Cellini & Cuccia, 2015; Cheng & Zhang, 2020).

Although previous studies provide theoretical references for assessing the cultural resilience of tourism destinations, there is no independent measurement of cultural resilience. The cultural resilience of tourism destinations is regarded as a subdimension or subsystem of socioecological and community resilience, and its subdivided evaluation or comprehensive measurement is conducted within these resilience systems (Bui et al., 2020; Cheer et al., 2019). For subdivision evaluation, divergent indicators have been utilized to measure cultural resilience within different resilience frameworks. For example, Hu et al. (2021) measured the cultural resilience of cultural heritage sites using indicators such as the grade of intangible culture heritage inheritors, protection value of village, and population density. Zhu et al. (2022) proposed a measurement model for ruralism-ecology system resilience in tourism destinations and assessed the cultural resilience by the number of cultural commercial streets, A-grade scenic spots, the number of clan ancestral halls, and folk museums. For comprehensive measurement, some scholars believe that cultural resilience is a component of rural resilience in tourism destinations but do not provide a subdivided measurement. Instead, they construct the above resilience evaluation framework from a comprehensive level, including dimensions of absorption, adaptability, transformation, redundancy, diversity, and multiscale connectivity. This framework is used to construct indicator layers from comprehensive statistical and scale data, such as locational conditions, confidence in tourism recovery, and community income diversity (Jia et al., 2022; Zhang et al., 2022).

However, the evaluation indicators of cultural resilience in different resilience frameworks show significant differences and disagreements. Considering the heterogeneity among the constituent elements of

cultural systems of different destinations (Yu & Xu, 2019), these objective statistical indicators may not be universally applicable (Haque & Hostetler, 2021). Resilience is characterized by complexity and dynamism (Prayag, 2023), making it difficult for objective statistical indicators to determine the actual scale and state of the cultural system's response to disturbances. In addition, the scale measurement needs to follow a rigorous evaluation development step, which has been neglected in related studies, leaving the practical utility of such evaluation systems questionable (dos Santos et al., 2024). Significantly, the predominant reliance of the above studies on objective statistical indicators to measure the cultural resilience of tourism destinations lacks a dissection of its components, leading to significant limitations in its academic understanding and theoretical comprehension. Importantly, a targeted evaluation tool for cultural resilience is essential for tourism destinations to monitor and assess their cultural systems' resilience state, enabling quick responses to ensure sustainable cultural development. Therefore, to address these research deficiencies, this study separates cultural resilience from other resilience frameworks of tourism destinations, specifically exploring its components and strictly following the scale development process to construct a highly applicable CRHS measurement tool.

2.4. Cultural protection and inheritance of heritage sites

Tourism development brings economic benefits, enriches cultural communication channels, and provides strong support for cultural resource protection and inheritance at heritage sites (Hu et al., 2021; Liu et al., 2022b). However, cultural heritage may more easily face various challenges in tourism development. Scholars have found that cultural heritage resources can be disturbed and threatened by issues such as climate change, shifts in the era, local population movement, and the intrusion of popular culture brought about by globalization (Larson et al., 2013; Li et al., 2022; Lin & Liu, 2018; Tolkach & Pratt, 2021). Therefore, existing studies primarily discuss problems of interference or strengthening measures in cultural protection and inheritance and provide countermeasures from different aspects. Examples include mining cultural authenticity elements, constructing hierarchical governance structures, and transforming tourism business models (Liu et al., 2022a; Park et al., 2019; Szromek et al., 2021; Wang et al., 2022). Others also suggest digital technology as an effective means of cultural heritage protection that can reduce some external interference (Bec et al., 2019; Mandić & Kennell, 2021). These studies primarily analyze the disturbance factors and coping measures in the cultural protection and inheritance of heritage sites in isolation without systematically and profoundly exploring the resilience issues behind cultural protection and inheritance.

Significantly, cultural heritage destinations have been transformed from static tourist attractions with conservation purposes to dynamic experience sites centered around revitalizing and inheriting traditional culture. They are chronically in a state of change and vulnerability to disturbances (Lee & Lee, 2019; Tolkach & Pratt, 2021). However, scholars have noted that some disturbances and changes may be vital opportunities for heritage sites to adjust or update their cultural development patterns. They could help cultural heritage sites adapt to environmental changes and promote sustainability (Canavan, 2016; Yang et al., 2016). These heritage sites are also considered to have cultural resilience (Tolkach & Pratt, 2021). In other words, cultural resilience offers new ideas for the cultural protection and inheritance of heritage sites (Bui et al., 2020; Hu et al., 2021). This characteristic also indicates the necessity of comprehensively considering the disturbances and coping processes of heritage sites, thereby deeply analyzing the underlying resilience logic. Nevertheless, the studies above on certain disturbances or protective measures have neglected the in-depth theoretical exploration of how cultural heritage can successfully cope with disturbances, which is not conducive to the sustainable development of heritage sites in disturbed environments. Therefore, this study notes the contribution of cultural resilience to cultural protection and inheritance at heritage sites and focuses on its cultural resilience for in-depth theoretical analysis.

3. Research design

3.1. Research case

China's world cultural heritage sites were chosen as the representative research case for the following prominent reasons. First, as of June 2023, China had 38 world cultural heritage sites, indicating that China entered a new stage of cultural heritage protection and inheritance in the deep integration of culture and tourism. These destinations face multiple challenges in achieving high-quality and sustainable development in a dynamic, changing environment (with various disturbances) (Hu et al., 2021). Second, the development and inheritance of these heritage sites have taken place in Chinese society, and their protection principles, development background, and supporting conditions have certain commonalities. These commonalities can minimize survey errors that may be caused by significant differences in national backgrounds in the exploration of cultural resilience. Third, China's world cultural heritage includes multiple types of historical cities, island communities, and mountain religions. Simultaneously investigating practices in different types of heritage tourism sites contributes to fully grasping the multisource information on the evolution of their cultural resilience and developing a scale with strong universality (Wang et al., 2023). On these grounds, China's world cultural heritage sites have strong adaptability and representation in this study.

3.2. Methods and design

Scale development is often based on Churchill Jr's (1979) research and has gradually evolved into increasingly rigorous implementation procedures. Despite some discussions and controversies surrounding Churchill Jr's (1979) steps, they remain a crucial reference for most scale development research (including non-tourism fields). Therefore, this study followed the procedures of Churchill Jr (1979), Rasoolimanesh et al. (2023), and other scholars to develop a CRHS scale (Otoo et al., 2021; Sun et al., 2020; Wang et al., 2023; Zhang et al., 2024). It was conducted through the following steps (Fig. 2):

Study 1: dimension exploration. Given the unclear structure of CRHS, a semi-structured interview outline was designed after determining its conceptual connotation, and interview texts were collected. Additionally, other multisource qualitative data were gathered. The dimensions of cultural resilience were extracted using grounded theory based on reference to the relevant literature.

Study 2: Scale development. The initial scale of cultural resilience was compiled. The initial questionnaire was evaluated and modified by experts, and the first round of the questionnaire survey was followed by exploratory factor analysis (EFA) and scale revision.

Study 3: Scale testing. The second questionnaire round was conducted after incorporating other related variables. Then, reliability and validity analysis, confirmatory factor analysis (CFA), nomological validity, and standard validity tests were carried out. Notably, this study combined field surveys and online surveys to collect data to prevent bias that may arise from single-channel data collection (Otoo et al., 2022).

4. Study 1: Dimension exploration

4.1. Data collection

This study collected qualitative data by field and online interviews, and the ancient city of Quanzhou in China was chosen as the field survey case site. It has a long history and rich heritage culture, with 22 subdivided cultural sites (Wang et al., 2023). Interviews were conducted with four types of stakeholders of heritage sites for the following

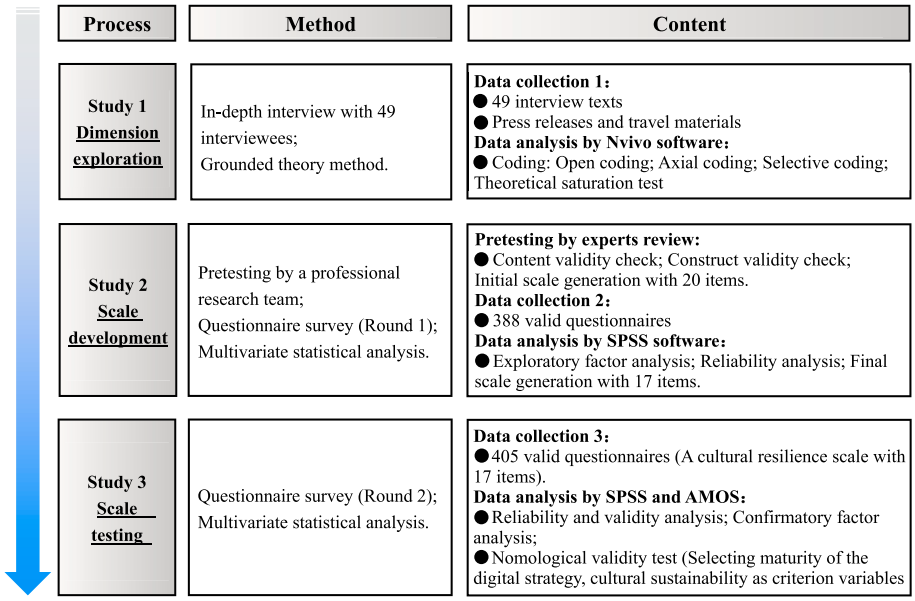

Fig. 2. Research design process.

reasons: (1) The construction of CRHS relies on the joint efforts of various stakeholders, such as the government, community, and tourism enterprises, rather than the actions of a single subject (Bui et al., 2020; Tsao & Ni, 2016). Therefore, some interviewees were included in the survey, including tourism site/enterprise workers and residents at heritage sites, local government staff (focusing on positions related to heritage culture), and tourists with in-depth cultural experience (Bui et al., 2020; Tsao & Ni, 2016). Among them, tourists were targeted towards groups with a strong interest in a particular cultural heritage site. They have repeatedly engaged in in-depth cultural experiences and tend to understand better its cultural development (Tsao & Ni, 2016). (2) Developing a scale for non-specified groups, interview data from multi-stakeholders can be used to generate items, and conducting surveys with multi-stakeholders can further strengthen triangulation (Wang et al., 2023; Zhang et al., 2024). Analyzing this multisource information also helps obtain a comprehensive view of their perspectives on how the cultural system successfully responds to disturbances. Moreover, the online interview recruited residents and tourists who understood the cultural development of China's cultural heritage sites from Sina Weibo (Wang et al., 2023). All recruits were over 18 years old. In addition, some respondents chose online interviews due to scheduling constraints.

A semi-structured interview outline was designed (Table 1). Initial questions were set to ensure that respondents understood the cultural development of heritage sites. Following this, questions focused on the core concepts of CRHS, such as cultural disturbances and their coping processes. These questions were developed with reference to resilience literature (Jiang et al., 2021; Taylor et al., 2022; Williams et al., 2020) and were reviewed and revised by cultural heritage experts. Table 1 summarized the interview questions that were progressively posed based on the degree of cultural understanding, cultural disturbances, coping processes, specific changes, and manifestations. Attentively, the interview was terminated to exclude the interviewee who could not answer midway and was not knowledgeable about the process of the cultural system of heritage sites in successfully responding to disturbances.

Based on the principles for representativeness and thematic relevance of respondents, interviewees were selected through purposive sampling and snowball sampling until the interviews reached theoretical saturation (Wang et al., 2023). The researchers conducted formal interviews from March to April 2023, starting from the cultural elements the interviewees knew best. They flexibly asked questions around the theme according to the specific answers from different interviewees, and all interviews averaged approximately 30–90 min. Seventeen residents, 8 cultural heritage and government workers, 4 cultural heritage inheritors, 13 operators and employees of cultural tourism enterprises, and 7 tourists were interviewed. In addition to tourists, most interviewees had dual identities: residents and cultural heritage workers. The respondents consisted of 23 males and 26 females, mainly aged between 26 and 44, and a majority of them had an educational level of college and above.

To ensure the saturation of cultural resilience data, news reports published by local authoritative newspapers and relevant policy documents issued by the government in China's cultural heritage sites were used as supplemental materials (Bui et al., 2020). Materials unrelated to the theme were screened out. Moreover, this study also captured travel data on Mafengwo using Python software (Wang et al., 2023), and "cultural heritage site name + culture + tourism experience" was used as the search keywords. After two types of data cleaning and screening, more than 30 news reports, policy documents, and 645 travel notes were obtained. These data underwent qualitative coding analysis, and a comparative analysis was then conducted to determine if new theoretical categories and logical relationships emerged, ensuring triangulation.

4.2. Data coding

4.2.1. Open coding

This study used grounded theory to code the interview data systematically, and the coding team consisted of 2 professors, 2 doctoral students, and 2 master's students. Open coding aimed to break down the raw data into different units of meaning. First, the 3 graduate students simultaneously conducted back-to-back coding around the research theme. They repeatedly analyzed the original interview texts, coding sentence by sentence, extracting critical information, and labeling it. Conceptualization and categorization were done through iterative comparisons and upward inductions until no new initial categories emerged in the data (Wang et al., 2023). Furthermore, the names of the corresponding concepts or categories were determined based on the original words in the interview texts. Subsequently, all members (led by 2 professors) discussed the initial coding results repeatedly, continuously modified contentious content and consulted external experts until

Table 1
Semistructured interview content.

Sample interview question	Interviewee	Purpose
⊙ Age, education level, occupation, length of residence (residents), length of work (workers in tourism attractions/ enterprises or government), number of visits (tourists), etc.	All respondents	Understanding respondents' basic information.
⊙ What cultures (e.g., architecture, food, folklore) do you think this heritage site includes? ⊙ Do these cultures still exist today? ⊙ How well have they been preserved or developed?	Residents, Government workers, Tourism attraction/ enterprise workers	Guiding respondents to introduce the cultural elements they know best and assess whether these cultural elements still exist.
⊙ What do you know about the culture here? ⊙ What culture have you experienced here? ⊙ What cultural elements appeal to you the most?	Tourists	
⊙ Could you tell us about the disturbances (threats) or obstacles (shocks) encountered in the development and inheritance of these cultures thus far? ⊙ How have you dealt with/ resolve these disturbances? ⊙ What is the specific process?	Residents, Government workers, Tourism attraction/ enterprise workers	Starting from the cultural elements familiar to the respondents, enabling them to express their understanding and views on cultural resilience, thereby collecting information about the basic elements and related practices of cultural resilience.
⊙ Please talk about how well these cultures are preserved or developed now based on your experience. ⊙ Do you have any suggestions for this?	Tourists	
⊙ Did these cultures change (compared to their original state) after encountering these disturbances (threats) or obstacles (shocks)? ⊙ In what ways?	Residents, Government worker, Tourism attraction/ enterprise workers	Continuing to dig deeper into the specific practical performance of cultural resilience.
⊙ Have you experienced the traditional/local characteristics of these cultures? ⊙ Please talk about whether they have noted have some innovations or changes.	Tourists	

Note: Respondents include residents (some residents are also cultural heritage inheritors), government workers, tourism attraction/enterprise workers, and tourists at cultural heritage sites.

a consensus was reached. After merging and deleting overlapping or invalid categories, 34 initial categories were identified (Appendix 1).

4.2.2. Axial coding

Axial coding regrouped the initial categories summarized by open coding. The coders continued to continuously compare and analyze the potential implicit structures among the initial categories, uncovering new concepts and categories by connecting these independent categories. They explored the internal logical relationships between them, and classified and aggregated the initial categories based on these associations to obtain the main categories. Subsequently, five main categories were identified by merging the related concepts that reflect the same theoretical issue (Table 2), cultural resistance, restoration, learning ability, adaptability, and growth.

4.2.3. Selective coding

Selective coding continued to explore the relationships and patterns among the main categories, leading to the theoretical abstraction that forms the core category and theoretical model. The coding team systematically integrated and refined the main categories, narrowing down the core research question and identifying 'cultural resilience of heritage sites' as the core category that governed all other categories, and constructed its conceptual model (Fig. 3). The complete storyline was: After the cultural system of a heritage site experiences various disturbances or shocks, it can adapt to contemporary society and the cultural tourism environment based on cultural resilience, continuing to evolve and achieve cultural heritage protection and inheritance. During this process, cultural resistance, restoration, learning ability, adaptability, and growth interconnect and provide feedback in the practice of cultural inheritance and development, promoting the formation of CRHS.

4.2.4. Theoretical saturation

To further verify the theoretical saturation, this study randomly selected 3/4 of the interview texts for the above coding, and the remaining 1/4 of the data was used to test the theoretical saturation. Comparative analyses revealed that no new significant constructs emerged (Wang et al., 2023). Moreover, a new round of programmatic coding analysis was conducted on these news reports, policy documents, and travel data, and the coding results obtained from the interview texts were triangulated for verification. Repeated comparisons did not reveal new categories and relationships (Bui et al., 2020; Sun et al., 2020). Therefore, the cultural resilience model constructed had reached theoretical saturation.

5. Study 2: Scale development and testing

5.1. Initial scale generation and purification

Based on the above coding results, the specific item formulation was compiled by extracting the content from the original interview text and referring to the relevant literature (see Appendix 3). The coding

Table 2
Axial coding results.

Main category/ dimension	Concept definition	Subcategory	Reference
Cultural resistance	During the development of heritage tourism, the ability of the cultural system of heritage sites to maintain the core structure and function to operate steadily by utilizing its conditions to resist shocks or disturbances. Its focus is mainly to prevent the cultural system of heritage sites from declining or collapsing after a shock.	Local cultural identity; cultural inertia formation; cultural development control; cultural charm attraction; cultural atmosphere infiltration; cultural social support	Melián-Alzola et al. (2020); Lindsay-Smith et al. (2022)
Cultural restoration	During the development of heritage tourism, the ability of the cultural system to recover or awaken its crucial functionalities after experiencing shocks or disturbances. These functionalities may not necessarily follow the original trajectory of the system but instead break the original path to achieve reactivation.	Cultural changing inheritance; cultural project restoration; cultural landscape restoration; cultural experience activation; cultural business operation restoration	Guo et al. (2018); Melián-Alzola et al. (2020)
Cultural learning ability	During the development of heritage tourism, the ability of the cultural system to continuously absorb various knowledge resources (e.g., traditional and emerging knowledge) and conduct corresponding exploration, research, and transformation, and application, thereby coping with shocks. Any disturbances or shocks that threaten cultural heritage need to be taken seriously, as they may provide opportunities for learning and exploration.	Cultural exchange and sharing; cultural continuous learning; cultural study tour statring; cultural knowledge explanation; cultural digital exploration	Guo et al. (2018); Lindsay-Smith et al. (2022)
Cultural adaptability	During the development of heritage tourism, the ability of the cultural system to successfully cope with disturbances or shocks through timely adjustments and transformations, such as shifts in the structure of internal elements. The aim is to break through the original cultural development model of the system and establish new development paths.	Multicultural coexistence; culture changes with the times; cultural integration into life; cultural adaptive maintenance	Pathak and Joshi (2021); Holladay and Powell (2013)
Cultural growth	During the development of heritage tourism, the ability of the cultural system to evolve to a better state or a higher level compared to the initial state through effective transformation and innovation after being affected by shocks or disturbances. The aim is to enhance the potential of the cultural system for its sustainable development.	Cultural corrective progress; cultural transformative enhancement; cultural element reorganization; cultural tourism product update	Chowdhury et al. (2019); Guo et al. (2018); Prayag and Dassanayake (2022)

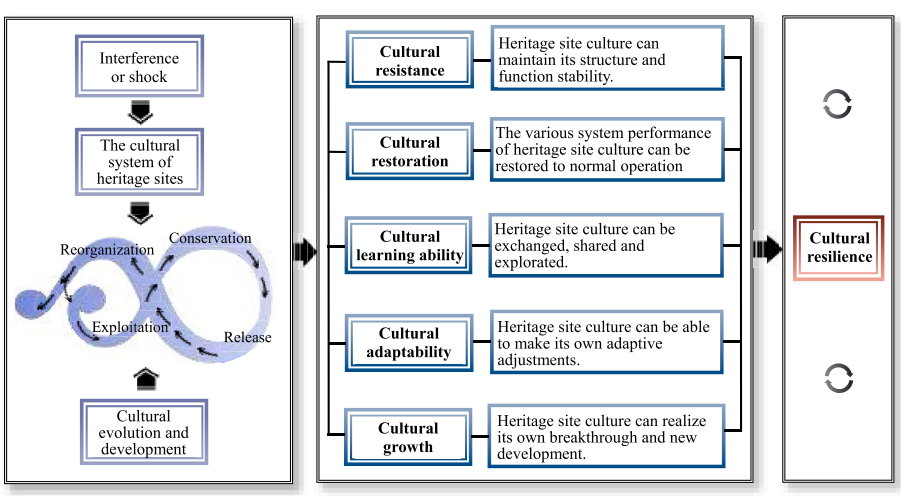

Fig. 3. Conceptual model of cultural resilience of heritage sites.

continuously compared the views of different stakeholders on CRHS and extracted information and concepts closely related to them. This process led to the formation of a scientifically rigorous scale with a comprehensive perspective. The initial scale consisted of 24 items, and the following measures were taken to ensure its face and content validity. First, our internal expert team (consisting of 3 experts and 3 doctoral candidates) conducted several rounds of evaluation and adjusted and optimized the items. The objective of this process was to prevent the following issues: insufficient matching between items and their construction, repetition between items, and abstract or academic expression of items that makes people difficult to understand (Wang et al., 2023). Second, an external research team was organized to evaluate the content validity of the CRHS scale after its initial items were developed. The team included 2 professors, 1 associate professor, 4 doctoral candidates, and 5 master's program students (studying the integrated development of culture and tourism, the culture in heritage sites). The first part of the evaluation materials introduced the concept connotation for CRHS, and the second part contained the main categories and their concept interpretation and the subcategories and corresponding items. The applicability and matching degree of items to their structure was evaluated by scoring, ranging from 1 (completely unsuitable) to 5 (completely suitable) (Sun et al., 2020). Two rounds of evaluation and repeated exchanges and discussions were conducted until all experts reached a consensus. Furthermore, four items were deleted, and the expression of the remaining items was revised based on the discussion results. Finally, 20 initial items were generated for the follow-up research.

5.2. Preinvestigation and scale purification

5.2.1. Questionnaire collection

This study conducted a small-scale field pretest in Quanzhou to evaluate the initial questionnaire's quality, and 51 valid questionnaires were collected by convenience sampling. According to the respondents' feedback, the item readability was adjusted and improved, and the final formal questionnaire with 20 items was obtained. Based on this, the first round of the questionnaire was designed. Specifically, the first part consisted of 20 items measured by a 7-point Likert scale ranging from 1 (completely disagree) to 7 (completely agree). The following reminder was set before all items: "Please answer all questions about the cultural heritage site you know best on the following basis: When the culture of this heritage site is disturbed or shocked (e.g., the COVID-19 pandemic), you think … " (Melian-Alzol et al., 2020). This setting ensured that the respondents completed the questionnaire smoothly. The second part included the sample statistical characteristics such as gender, age, and education.

The first round of questionnaire distribution combined a field survey (the ancient city of Quanzhou) and an online survey (Sojump) in May 2023. 388 valid questionnaires were obtained (Nunnally, 1967). Sixty invalid questionnaires with random or incomplete answers were deleted, with an effective rate of 86.61%. All the sample statistics are shown in Appendix 2.

5.2.2. Exploratory factor analysis

This study used SPSS software to perform exploratory factor analysis (EFA) to test the dimensional composition of the CRHS scale obtained from the above qualitative analysis and to purify the corresponding items. The analysis revealed a Kaiser–Meyer–Olkin (KMO) of 0.940 and a significant Bartlett sphericity test result, which suggested that EFA was appropriate (Chi & Chi, 2022). Furthermore, principal component analysis with a factor extraction criterion of an eigenvalue greater than 1 was selected. Then, we conducted an orthogonal rotation analysis based on the maximum variance method and deleted the following items based on the results (Appendix 4): (1) factor commonality is less than 0.5, (2) factor loading is less than 0.5, and (3) cross-load occurs in the same item, and its value exceeds 0.4 (Chi & Chi, 2022). Three items were deleted: *"CRS2: People habitually participate in cultural activities (e.g., folklore activities)," "CLA2: This destination continuously learns various cultural knowledge,"* and *"CG3: Cultural elements are integrated and reorganized in new ways"*. Finally, 5 dimensions were obtained with 17 items.

The Cronbach's alpha values for the five dimensions were all greater than 0.8 (0.835–0.942), with all factor loadings between 0.684 and 0.849 (>0.6), and the commonalities ranged from 0.661 to 0.912 (>0.6), explaining 79.03% of the total variance. All indicators met the standard (Chi & Chi, 2022; Sun et al., 2020). A scale comprising 5 factors and 17 items was obtained by EFA, including cultural resistance (4 items), restoration (4 items), learning ability (3 items), adaptability (3 items), and growth (3 items).

5.3. Formal investigation and scale test

5.3.1. Questionnaire distribution and analysis

Based on the above analysis, a second-round questionnaire was designed using 17 retained items on cultural resilience and predictive variables (i.e., maturity of the digital strategy and cultural sustainability) for subsequent nomological and predictive validity tests. The questionnaires were distributed in mid-June 2023 by combining an online survey (Sojump) and an offline survey (at the ancient city of Quanzhou). We collected 405 valid data points and excluded 42 invalid data points with an effective recovery rate of 90.60%, and the sample size reached the standard (Nunnally, 1967). The demographic characteristics of all respondents are detailed in Appendix 2.

5.3.2. Confirmatory factor analysis

This study further corroborated the underlying structure of cultural resilience found by EFA through confirmatory factor analysis (CFA). The results were shown in Table 3. The Cronbach's alpha values for the whole scale and each dimension were greater than 0.8, and all factor

Table 3
Descriptive statistics and confirmatory factor analysis.

Constructs and factors	Mean	Factor Loading	CR	AVE
Cultural resilience (*Cronbach's alpha = 0.958*)				
Cultural resistance (*Cronbach's alpha = 0.934*)			0.934	0.781
The culture is still liked and recognized by people	5.121	0.904		
There are management measures to ensure cultural development	4.921	0.871		
The culture can maintain enough attraction for people	5.084	0.872		
The strong local culture and living atmosphere are infectious	5.202	0.887		
Cultural restoration (*Cronbach's alpha = 0.919*)			0.919	0.740
The culture is still passed down from generation to generation	5.042	0.882		
The cultural programs are revived	4.889	0.838		
The cultural landscapes are restored according to their original appearances	4.988	0.866		
The cultural experiences are revitalized	4.946	0.855		
Cultural learning ability (*Cronbach's alpha = 0.879*)			0.879	0.709
Cultural information and knowledge are exchanged and shared	4.773	0.874		
Explanations are arranged to help people understand the culture	4.706	0.825		
This destination learns to use digital technology to preserve and develop the culture	4.716	0.824		
Cultural adaptability (*Cronbach's alpha = 0.891*)				
The cultural development changes in response to the environment	4.758	0.782	0.892	0.734
The culture is still integrated into the daily lives of local people	4.983	0.901		
Cultural preservation and development strategies are adapted to the environment	4.926	0.882		
Cultural growth (*Cronbach's alpha = 0.888*)				
The backwardness of the culture is adjusted while preserving its essence	4.896	0.857	0.891	0.732
The culture is improved and developed in line with the times	4.916	0.907		
Cultural tourism products are updated and upgraded with new elements	4.884	0.799		
Maturity of the digital strategy (*Cronbach's alpha = 0.889*)			0.889	0.668
Digital technologies are used to develop the culture	4.699	0.821		
Digital technologies for the cultural development are continuously enhanced	4.835	0.800		
Digital technologies for cultural development are evaluated and adapted	4.746	0.851		
Digital technologies are helpful for the development of cultural tourism	4.916	0.796		
Cultural sustainability (*Cronbach's alpha = 0.927*)			0.927	0.717
The culture is well preserved/passed on	5.123	0.869		
The culture is well developed	5.089	0.864		
There are abundant cultural activities	5.025	0.825		
The cultural heritage is well protected	5.069	0.849		
The culture is easy to understand and accept	5.096	0.826		

Note: N = 405.

loadings exceeded 0.7. Furthermore, the CR (composite reliability) for all dimensions was greater than 0.7, and the AVE (average variance extraction) was above 0.5. Moreover, Appendix 5 presented that the square roots of AVE (0.842–0.884) were greater than the correlation coefficients between these dimensions (0.475–0.589). All the above indicators met the required standards, demonstrating that our developed CRHS scale had good reliability, convergent validity, and discriminant validity (Chi & Chi, 2022; Sun et al., 2020).

This study determined the optimal factor structure for CRHS by constructing four competing models (Chi & Chi, 2022; Otoo et al., 2021). Model 1 comprised a first-order factor model of 17 indicators, Model 2 exhibited an uncorrelated five-factor model, Model 3 was a correlated five-factor model, and Model 4 represented a second-order model consisting of five sub-dimensions of CRHS. As shown in Table 4, the fit indices for Models 1 and 2 were substandard, while Models 3 and 4 had good and similar fit indices, with factor loadings greater than 0.7. Therefore, Models 3 and 4 presented the optimal factor structure of CRHS, indicating their suitability for second-order dimensionality measurement (Zhang et al., 2024).

5.4. Nomological and predictive validity

This study examined the theoretical relationships between cultural resilience and other related concepts to verify the nomological and predictive validity of the CRHS scale (Chi & Chi, 2022; Otoo et al., 2022). The maturity of the digital strategy emphasizes the perfection and stability of digital development strategies for cultural tourism in heritage sites (Forliano et al., 2023). Their cultural systems are vulnerable to various internal and external disturbances or shocks, and digital strategies can be used to effectively cope with these disturbances through cultural storage, transformation, and dissemination (Beel et al., 2017; Choi et al., 2021; Forliano et al., 2023). Specifically, digital strategies help identify disturbances in time and integrate multiple subjects through smart governance (Pesce et al., 2019). It also can virtually reconstruct and restore cultural heritage and minimize the depletion of cultural relics through alternative immersion experiences (Bec et al., 2019; Choi et al., 2021). More importantly, a mature digital strategy will help effectively avoid problems of mindlessly following trends and disconnecting between the construction and use in the digital development of heritage tourism (Forliano et al., 2023), which will help enhance cultural resilience. Furthermore, higher cultural resilience indicates that the cultural system of heritage sites exhibits greater capability for successfully coping with various perturbations. Cultural resilience can promote cultural protection, inheritance, and sustainable development in a constantly changing environment (Bui et al., 2020; Clarke & Mayer, 2017; Hu et al., 2021; Zhu et al., 2022). Therefore, this study posits that the maturity of the digital strategy can enhance cultural resilience, while cultural resilience can enhance cultural sustainability, and constructs a corresponding theoretical model (Fig. 4).

Table 4
Model comparisons of CRHS.

Fit index	Model 1 (First-order factor model)	Model 2 (5-order factor model with no correlation)	Model 3 (5-order factor model with a correlation)	Model 4 (Second-order model)
χ^2/df	16.944	8.664	1.758	1.811
RMSEA	0.199	0.138	0.043	0.045
CFI	0.654	0.834	0.947	0.983
GFI	0.563	0.712	0.947	0.944
AGFI	0.438	0.630	0.926	0.924
NFI	0.641	0.816	0.966	0.963
RFI	0.590	0.790	0.957	0.956
IFI	0.655	0.834	0.985	0.983
TLL	0.604	0.810	0.981	0.980
p-value	0.000	0.000	0.000	0.000

Note: ***P < 0.001.

Fig. 4. Theoretical model.

The items for maturity of the digital strategy were adapted from Forliano et al. (2023) and for cultural sustainability from Apak and Gurbuz (2023) (Table 3). They are suitable for testing the nomological and predictive validity of the cultural resilience scale. Items for these two variables were added in the second round of questionnaire collection to test the measurement model. The results showed that the correlation coefficients among the three variables were between 0.548 and 0.717 ($p < 0.01$), and the VIF was 1.754, indicating no multicollinearity problem. The CFA results of confirmatory factor analysis are shown in Table 3, with the corresponding indicators meeting the standards (Chi & Chi, 2022). Moreover, the structural equation model analysis by AMOS 22.0 demonstrated that maturity of the digital strategy significantly and positively affected cultural resilience ($\beta = 0.714$, $p < 0.001$), and cultural resilience had a significant positive impact on cultural sustainability ($\beta = 0.767$, $p < 0.001$). The model fit indices revealed that the model was reliable, and the relationship between variables was established (χ^2/df = 1.960; RMSEA = 0.049; CFI = 0.951; GFI = 0.951; AGFI = 0.931; NFI = 0.962; RFI = 0.954; IFI = 0.981; TLL = 0.977). Thus, the cultural resilience scale had good nomological and predictive validity (Chi & Chi, 2022).

Overall, this study followed a rigorous scale development procedure to develop and validate the CRHS scale. An initial scale with 24 items was constructed based on the coding results of grounded theory, and we deleted 4 and 3 items sequentially through content validity analysis and EFA. Subsequently, 17 items remained after CFA. In addition, the scale underwent multiple rounds of evaluation and adjustment by experts and respondents during its development and validation. Nomological and predictive validity tests were also conducted. These multiple qualitative and quantitative statistical analyses can confirm the content and structural validity. Ultimately, the CRHS was comprised of 5 dimensions and 17 items.

6. Discussion

6.1. Conclusions

First, this study proposes the concept of cultural resilience of heritage sites (CRHS) and establishes and validates a five-dimensional theoretical structure of CRHS. Based on the concepts of cultural resilience and evolutionary resilience, we focus on the notion of CRHS, elucidating its triggering situation (various disturbances during heritage tourism development), disturbed subject (cultural system), specific composition of the culture (tangible and intangible heritage elements), and manifestation (survival, recovery, adaptation, and continuous renewal and development). Although relevant research has attempted to develop cultural resilience into a clear and valuable concept, it aims to address how communities respond to various disturbances or changes (Beel et al., 2017; Clarke & Mayer, 2017). They offer limited theoretical guidance for achieving sustainable cultural heritage development in disturbed environments. In contrast, our proposed CRHS emphasizes a series of processes by which the cultural system responds to disturbances during heritage tourism development. It can directly address the dilemmas faced by the cultural system and is crucial for sustainable cultural development.

Additionally, our work refines the five theoretical dimensions of CRHS: cultural resistance, restoration, learning ability, adaptability, and growth. This structure extends the previous community resilience frameworks (Davis et al., 2021; Jia et al., 2022) and further newly

identifies cultural learning ability as a unique attribute, distinguishing the cultural system of heritage sites from traditional community or social-ecological systems. Community resilience involves multiple subsystems, such as ecology, economy, culture, and environment (Zhu et al., 2022). The exploration of its structural dimensions tends to comprehensively consider the overall performance or form of these subsystems in successfully responding to disturbances, potentially overlooking the unique characteristics of a particular resilience subsystem. Regarding the cultural systems of heritage sites, they often exist in a dynamically changing environment during the development of tourism and societal evolution. Its successful response to disturbances and continuous inheritance is primarily due to the ongoing absorption and exploration of diverse and emerging knowledge and information about the cultural system. Consequently, the system develops a strong, up-to-date cultural learning ability (e.g., the exploration of digital technology), which helps them quickly adapt to and effectively cope with various shocks or changes (Bui et al., 2020; Haque & Hostetler, 2021). Therefore, as a crucial sub-dimension of CRHS, cultural learning ability cannot be ignored.

Second, this study targets developing and validating a five-dimensional assessment scale for cultural resilience of heritage sites. We construct and quantitatively test a CRHS scale comprising cultural resistance, restoration, learning ability, adaptability, and growth. Its evaluation revolves around multi-stakeholders. In contrast, cultural resilience within the resilience framework of tourism destination systems is often viewed as a secondary subsystem, and objective statistical indicator layers are built (Hu et al., 2021; Zhu et al., 2022). These approaches face significant challenges in the universality of cultural resilience assessment systems. For example, the cultural resilience assessment indicators within the resilience frameworks of different tourism destinations are obviously divergent, and the cultural heritage elements they involve are difficult to cover multiple types comprehensively. Moreover, the design of adding some indicators assessed by scales has shown inadequacy in the scientific rigor of the development process (dos Santos et al., 2024; Jia et al., 2022). Therefore, this paper proposes to use scale items instead of statistical indicators to measure CRHS and strictly follow the scale development process to construct the corresponding measurement standards. It develops a five-dimensional scale containing 17 items with good reliability and validity, which provides relatively higher universality.

6.2. Theoretical contributions

First, our study first highlights cultural resilience of heritage sites and clarifies the resilience logic of cultural protection and inheritance in heritage sites. Previous studies provide theoretical guidance for the cultural protection and inheritance of heritage sites by clarifying some disturbances or shocks in their development and optimizing strategies (Bec et al., 2019; Lin & Liu, 2018; Tolkach & Pratt, 2021). Although some researchers argue that some disturbances may be significant opportunities for cultural protection and sustainable development, resilience thinking can effectively solve disturbances in the cultural system of heritage sites and help these sites adapt to changes and continue to move toward better development (Bui et al., 2020; Canavan, 2016; Hu et al., 2021). Neglectfully, current research has less theoretical exploration of cultural resilience of heritage sites, making it difficult to provide resilience solutions for their cultural protection and inheritance. In response, this study systematically deconstructs cultural resilience of heritage sites, deeply explores its theoretical logic, and innovates the theoretical ideas and viewpoints for cultural protection and inheritance research. The analysis of cultural resilience also provides theoretical guidance for the sustainable cultural development of heritage sites. Furthermore, it deepens the academic understanding of the theoretical relationship between disturbances and coping logic in cultural protection and inheritance.

Second, the findings clarify the structural dimension for cultural resilience of heritage sites and complete the independent dissection of the complex structure for the cultural resilience of tourism destinations. Cultural resilience has been incorporated into community and social-ecological resilience framework of tourism destinations to construct indicator systems (Hu et al., 2021; Zhang et al., 2022; Zhu et al., 2022). Some scholars propose that cultural resilience is an important subsystem of rural community resilience or cultural heritage resilience and that it includes dimensions such as resistance, adaptability, and transformation, as determined by a comprehensive analysis (Davis et al., 2021; Jia et al., 2022; Maurischa et al., 2023). Nevertheless, they have not identified the structural dimensions of tourism destination cultural resilience alone. Significantly, this study regards cultural resilience of heritage sites as an independent and complex system (Clarke & Mayer, 2017) and extracts its specific components of cultural resistance, restoration, learning ability, adaptability, and growth by grounded theory. A more precise conceptual connotation and extension for the CRHS can be grasped in doing so. It also strengthens the theoretical interpretation of its essential connotation and further deepens the in-depth development of cultural resilience theory.

Third, this work develops an independent scale of cultural resilience of heritage sites and expands the theoretical system for evaluating tourism destination resilience. Existing studies center more on the evaluation index systems around community and social-ecological resilience in different types of tourism destinations. Cultural resilience is considered one of their subdimensions or subsystems to measure (Hu et al., 2021; Zhu et al., 2022). However, the measurement indicators of cultural resilience within the resilience system of different types of tourism destinations differ significantly. They are less universal, making it difficult to directly measure the cultural resilience of tourism destinations (Haque & Hostetler, 2021). Accordingly, this study separates cultural resilience from the previous tourism destination resilience framework, conducts research based on multiple cases, and follows the scale development process. The purpose is to build and verify an independent and universal measurement tool for cultural resilience of heritage sites (Clarke & Mayer, 2017). This exploration extends the system of content for tourism destination resilience research around the quantitative evaluation level and refines and expands its cultural perspective. In addition, the findings lay a foundation for subsequent quantitative studies on the cultural resilience of tourism destinations. It can also deepen the theoretical framework for cultural supervision and governance of heritage sites.

6.3. Management implications

This study provides practical guidance for assessing and strengthening cultural resilience in heritage sites for multiple subjects of the government, community, and enterprise.

To assess cultural resilience, first, the government and community managers of heritage sites can use the cultural resilience scale to regularly assess the cultural development status and dynamically adjust their cultural disturbance or risk response measures based on the CRHS dimension. This process contributes to preventing the cultural system from falling into development imbalance or collapse. Second, the subdimensions of cultural resilience are evaluated in detail through this scale, which helps adopt more reasonable plans and clarify priorities at different stages of responding to disturbances or shocks (Choi et al., 2021). This leads to efficient and differentiated practices of cultural resistance, restoration, learning ability, adaptability, and growth. Finally, the staff of tourism enterprises can use the scale to evaluate various indicators for cultural resilience of heritage sites. Consequently, it will be easy to understand the potential hidden dangers in consumers' cultural experiences, thereby formulating effective interference and risk prevention plans for the production and operation of cultural tourism products and revising them promptly.

To strengthen cultural resilience, first, heritage sites need to build

resilience based on local cultural characteristics, focusing on the creation of a strong local cultural atmosphere and the excavation and display of unique cultural charm. It is also important to fully integrate these practices into the daily lives of destination residents (Hu et al., 2021). Second, ensuring the rapid recovery of tourism culture from the impact is essential, including cultural landscape restoration and experience relaunching (Hu et al., 2021). For example, the government and scenic spots can attract traditional craftsmen to restore ancient buildings by making existing materials look like old ones. Third, the key point of cultural learning lies in exploring digital technological knowledge, such as using digital multimedia to store cultural resources and creating cultural immersion experience products (Choi et al., 2021). In addition, the government and tourism enterprises should provide long-term learning channels to enhance understanding of traditional cultural characteristics. Specifically, community cultural exchanges, cultural explanation manuals (paper + electronic), and online classes for intangible cultural heritage mobile classes could be created. Fourth, tourism enterprise workers and cultural inheritors need to have the flexibility to adjust and transform the presentation and operation mode of cultural tourism products according to the type of disturbance (Bui et al., 2020; Sakakibara, 2017). For example, they can expand online viewing channels for traditional cultural performances and adapt traditional dramas to incorporate modern pop culture elements. Finally, the secret of cultural growth is to "remove the bad and keep the good". Communities and tourism enterprises should adjust or even remove the backward elements of traditional culture and retain the essence when developing cultural tourism products or organizing traditional cultural activities (Holtorf, 2018).

6.4. Limitations and future prospects

This study has strong theoretical and practical significance but also some limitations. The scale development focuses on China's world cultural heritage sites, which may be limited by sampling during the investigation. Future research could explore cultural resilience in different cultural backgrounds, destination types, and disciplinary systems, test the universality of the scale (Sun et al., 2020), and extend the theoretical application of the cultural resilience scale (Hu et al., 2021). Additionally, this study developed the CRHS scale to lay a foundation for subsequent empirical research. Remarkably, cultural resilience is susceptible to multiple factors during the development of heritage sites (Hu et al., 2021). Future studies can further analyze the influencing factors and dynamic evolution mechanisms for cultural resilience of heritage sites and expand the theoretical system of cultural resilience of tourism destinations. Moreover, this work finds that digital technology can empower cultural learning in heritage sites. However, some studies have suggested that digital empowerment may be a double-edged sword for cultural heritage protection and inheritance (Bec et al., 2019). Therefore, future studies can further investigate the influence boundaries and the logic of digital empowerment on cultural resilience of heritage sites. Finally, the widely applied procedures by Churchill Jr (1979) may harbor some judgments and controversies. Therefore, this paper synthesizes Churchill Jr's (1979) approach with other scholars' adapted and upgraded scale development procedures. For example, we have distributed multiple rounds of questionnaires (the sample data was representative and the sample size was sufficient) and tested nomological validity. This way contributes to improving the scientificity and rigor. However, some contentious viewpoints may persist, as different scholars hold varying opinions. For instance, Churchill Jr's (1979) procedure has yet to adequately identify the nature of the dimensions, e.g., reflective vs. formative (Rasoolimanesh et al., 2023). Future research could further explore and validate the controversies and relatively other procedures.

CRediT authorship contribution statement

Rui Li: Writing – original draft, Investigation, Formal analysis. **Yong-Quan Li:** Supervision, Project administration, Funding acquisition, Conceptualization. **Wen-Qi Ruan:** Writing – review & editing, Validation, Supervision, Conceptualization. **Shu-Ning Zhang:** Validation, Methodology. **Mei-Yu Wang:** Investigation, Data curation.

Declarations of interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

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Appendix 1. Open coding example

Initial category	Initial concept	Examples of primary materials
Love for local culture	Tourists enjoy local traditional culture	Just like tourists from Beijing, Shanghai, and Guangzhou, they are very fond of our local traditional culture and are still quite passionate about it. (MT-22-6)
	Local people love traditional folklore	We eat rice crackers on the Qingming Festival, adults and children like to eat them, so we continue to do so according to the custom. (WJ-23-2)
	Residents continue to love local culture	The community residents have a spirit of persistence, and a true and pure love for these cultures. (HH-6-20)
Cultural atmosphere edification	Impress young people by what they see and hear	Young people are influenced from a young age as we grow up in a cultural environment and atmosphere surrounded by traditional customs, which have continuously impacted us. (WY-44-8)
	Intergenerational infection of family beliefs	If parents have the faith, then they can lead their children to participate in some relevant cultural activities when they are young. The children can receive the teaching of words and deeds or intergenerational infection from their families in this process. (HH-6-7)
	Strong atmosphere of faith	There are many temples, and we felt the strong atmosphere of faith, just like Quanzhou in China is known as the 'Office of the Gods'. (XP-48-1)
Cultural landscape restoration	Timely repair of ancient buildings	Property staff often take a look around the scenic spots, because there may be bumps in places with many tourists. Once something goes wrong, they promptly go to repair it. (AL-9-11)
	Antique repair and renovation	These traditional building have been renovated in an antique style. (SY-41-7)
Launch of cultural study tours	Integration of intangible cultural heritage into study tours	We launched this study tour of puppet making, in which lectures and handcrafts are combined together, so that children can get a deep understanding of these cultures. (MT-21-7)
	Combination of folk culture and study tours	Now we deepen people's understanding of some folk culture through study tours, not only pure worship but also some folk activities can be combined with study tours. (XY-32-22)
Local Cultural change	Changes in the rules of traditional festivals	Now the rules and size of Moon-cake gambling (a traditional activity of the Chinese Mid-Autumn Festival) can be changed according to different people, and you can set it by yourself. (YT-38-17)

(*continued on next page*)

(*continued*)

Initial category	Initial concept	Examples of primary materials
	Changes in traditional food fillings	The fillings of the run cake (a traditional food eaten during the Chinese Tomb Sweeping Festival) have been evolving, as individual tastes vary. (KF-36-2)
	Changes in the form of folk activities	The tower burning (a Chinese traditional folk activity) gradually became a collective activities, but it started out as a small kind of fire outside the house, and now the individual cannot burn it outside. (JW-28-3)
	Decoration of ancient buildings for the needs of life	Because the room of the tulou is relatively small, some residents enlarge the room by knocking through between the two small rooms and then properly decorate it by themselves. (GJ-35-3)
Cultural adjustment progress	Traditional culture discards the bad and extracts the good	These traditional cultures has extracted the essence and discarded the dross, for example, the parts of Zhu Xi's culture we receive today are the better parts. (LL-40-4)
	Replacement of traditional festival diet habits	In the past, we had to eat cold food on the Chinese Tomb Sweeping Festival, that is, we could not eat hot food, and now it goes without this requirement. (YF-17-2)

Note: The source material is marked sentence by sentence as "interviewee + text number + sentence number".

Appendix 2. Characteristics of the research sample

Variables	Round 1 (388)		Round 2 (405)		Variables	Round 1 (388)		Round 2 (405)	
	Frequency	Percentage (%)	Variable	Percentage (%)		Frequency	Percentage (%)	Frequency	Percentage (%)
Gender					**Average monthly income (¥)**				
Male	201	51.8	190	46.9	≤3500	63	16.2	91	22.5
Female	187	48.2	215	53.1	3501–5000	66	17.0	52	12.8
Age					5001–8000	98	25.3	86	21.2
18–25	87	22.4	87	21.5	8001–12500	99	25.5	95	23.5
26–34	115	29.6	78	19.3	12501-38500	42	10.8	70	17.3
35–44	101	26.0	144	35.5	≥38501	20	5.2	11	2.7
45–54	41	10.6	57	14.1	**Level of cultural understanding**				
55–64	39	10.1	37	9.1	General	126	32.5	107	26.4
65 and above	5	1.3	2	0.5	Relatively good	194	50.0	231	57.0
Educational level					Excellent	68	17.5	67	16.6
Below high school	24	6.2	14	3.4	**Role**				
High school/ secondary school	53	13.6	72	17.8	Resident	218	56.2	243	60.0
College	97	25.0	109	26.9	Tourist	97	25.0	66	16.3
Undergraduate	173	44.6	164	40.5	Tourism enterprise worker	34	8.8	40	9.8
Master and above	41	10.6	46	11.4	Non-local student or employee of non-cultural tourism enterprises	13	3.3	35	8.6
Occupation					Others	26	6.7	21	5.3
Enterprise employee	101	26.0	105	25.9	**Length of residence (years)**				
Civil servant	31	8.0	26	6.4	≤5	14	6.4	28	11.5
Educational researcher	37	9.5	47	11.6	6–10	35	16.1	49	20.2
Individual operators	53	13.7	47	11.6	11–20	48	22.0	69	28.4
Students at school	42	10.8	41	10.1	≥21	121	55.5	97	39.9
Technicians	38	9.8	38	9.4	**Number of visits (times)**				
Freelancers	28	7.2	51	12.6	2–3	65	67.0	43	68.3
Leavers/retirees	10	2.6	2	0.5	4–5	18	18.6	13	20.7
Others	48	12.4	48	11.9	≥6	14	14.4	10	16.0

Appendix 3. Example of sources for item generation

Interviewee	Interview content	Initial items
The manager of an intangible cultural heritage theater	Just like tourists from Beijing, Shanghai, and Guangzhou, they are very fond of our local traditional culture and are still quite passionate about it. (MT-22-6)	The culture is still liked and recognized by people
Resident	We have a heartfelt love for our indigenous culture, which is the source of everything, so these customs have been passed down. (HY-37-21)	
Resident	Young people are influenced from a young age as we grow up in a cultural environment and atmosphere surrounded by traditional customs, which have continuously impacted us. (WY-44-8)	The strong local culture and living atmosphere are infectious
Tourist	There are many temples, and we felt the strong atmosphere of faith, just like Quanzhou in China is known as the 'Office of the Gods'. (XP-48-1)	
Tourist attraction workers	We renovated these old buildings in a traditional way, preserving the architectural features, and we have specially invited those old craftsmen to restore the ancient houses or repair the mortise and tenon structures. (YF-17-23)	The cultural landscapes are restored according to their original appearances
Government worker	These traditional buildings have been renovated in an antique style. (SY-41-7)	
Resident	The process of traditional customs has been simplified a lot, and the necessary items are easier to prepare, often bought from stores, reflecting that they are evolving with the times. (ZX-29-3)	The cultural development changes in response to the environment
Resident	The flavor of the four-fruit soup (a traditional food) has changed quite a lot, even up to eight additional ingredients, such as jelly and QQ candy, have been included. (YY-36-2)	

(*continued on next page*)

(*continued*)

Interviewee	Interview content	Initial items
Resident	These traditional cultures have extracted the essence and discarded the dross; for example, the parts of Zhu Xi's culture we receive today are the better parts. (LL-40-4)	The backwardness of the culture is adjusted while preserving its essence
Resident	In the past, women were not allowed to sweep the tombs during the Tomb Sweeping Day, but now women can freely participate in these traditional folk activities. (YF-17-3)	

Appendix 4. Exploratory factor analysis results of CRHS (Before deleting the items)

Constructs	Items	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
Cultural resistance	CRS1	0.810				
	CRS2	0.489				
	CRS3	0.739				
	CRS4	0.841				
	CRS5	0.751				
Cultural restoration	CRO1				0.701	
	CRO2				0.738	
	CRO3				0.793	
	CRO4				0.735	
Cultural learning ability	CLA1			0.694		
	CLA2		0.404	0.698		
	CLA3			0.769		
	CLA4			0.815		
Cultural adaptability	CA1					0.783
	CA2					0.803
	CA3					0.785
Cultural growth	CG1		0.812			
	CG2		0.837			
	CG3		0.600	0.605		
	CG4		0.823			
Cronbach's alpha	0.947	0.870	0.880	0.885	0.878	0.924
Total variance of interpretation (%)			75.769			

Appendix 5. Discriminant validity analysis

Constructs	*Mean*	*S.D.*	1.	2.	3.	4.	5.
1. Cultural resilience	5.082	1.573	**0.884**				
2. Cultural restoration	4.966	1.529	0.587	**0.860**			
3. Cultural learning ability	4.732	1.538	0.475	0.585	**0.842**		
4. Cultural adaptability	4.889	1.577	0.589	0.548	0.534	**0.857**	
5. Cultural growth	4.899	1.548	0.546	0.532	0.516	0.568	**0.856**

Note: The value below the diagonal is the correlation between structures. All correlations are at $p < 0.01$ (**$p < 0.01$). The square root of average variance extraction is shown on the diagonal in bold.

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